

Communication Protocol Specification

Operation: Send Information

Allows the server to send information to clients.

Response: Server → Clients

Field	Size	Description
Type	1 Byte	Packet Type (1: packet)
Number	4 Bytes	The packet sequence number
Size	4 Bytes	Total number of characters in the checksum
Information	1 Byte	checksum
Number	4 Bytes	Total number of packets
Number	4 Bytes	Packet number
Number	2 Byte	Number layer
Size	4 Bytes	Size of the data to be sent
Information	Variable	Data to be sent

Allows the client to send information to server. **Request:** Client → Server

Field	Size	Description
Type	1 Byte	Packet Type (1: packet, 0: Register)
Number	4 Bytes	The packet sequence number
Size	4 Bytes	Total number of characters in the checksum
Information	1 Byte	checksum
Number	4 Bytes	Total number of packets
Number	4 Bytes	Packet number
Number	2 Byte	Number layer
Size	4 Bytes	Size of the data to be sent
Information	Variable	Data to be sent

Operation: ACK

Response: Client → Server

Field	Size	Description
Type	1 Byte	Packet Type (2: ACK)
Number	4 Bytes	The packet sequence number

Operation: NACK

Response: Client → Server

Field	Size	Description
Type	1 Byte	Packet Type (3: NACK)
Number	4 Bytes	The packet sequence number

Usage Examples

Register Client

```
0
0
```

Listing 1: Examples of Register Client Packets to Server

Send Information

```
100010001CHECKSUM00050001010004HOLA
100020001CHECKSUM00050002010005PABLO
100030001CHECKSUM00050003010004HOLA
100040001CHECKSUM00050004010004Juan
100050001CHECKSUM00050005010011Informacion
```

Listing 2: Examples of information sending packets

ACK

```
20001
20002
```

Listing 3: Examples of ACK packets

NACK

```
30001
30002
```

Listing 4: Examples of NACK packets